SpaceX Compliance: licenses, spectrum, insurance, and non-English primary law

Search Audit · 100 prompts · 6 providers · run 2026-10-09

Parallel gives SpaceX the best balance of accuracy, cost and speed. 4 providers tie on accuracy, so cost and speed decide.

MOST CORRECT

Parallel: 61 of 100

Tied within chance with Exa, Grok native and Keenable router.

CHEAPEST OF THE MOST CORRECT

Parallel: $0.440 per prompt

61 of 100 correct.

FASTEST

Exa: 43.9s median

60 of 100 correct.

Ranking

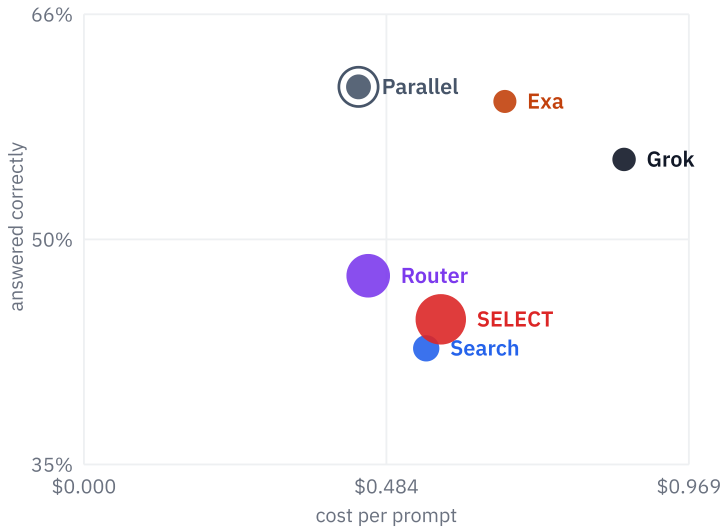
Value score combines accuracy, cost and speed per prompt (wrong = 0). Weights: accuracy 60, cost 25, speed 15.

#	Provider	Value score	Correct	$ / prompt	Time
1	Parallel	65	61/100 tied	$0.440	56.7s p90 5.1m
2	Exa	63	60/100 tied	$0.674	43.9s p90 5.0m
3	Grok native	57	56/100 tied	$0.865	44.5s p90 4.8m
4	Keenable router	55	48/100 tied	~$0.455	3.6m p90 12.4m
5	Keenable search	53	43/100	~$0.548	1.2m p90 9.0m
6	Keenable SELECT	50	45/100	~$0.571	4.6m p90 15.5m

green marks the best on this table. A tie means the accuracy gap is within chance (paired McNemar test, Holm-corrected, vs Parallel). ~ = priced from a list price.

Accuracy vs cost

Up and left is better. Dot size = median time. Ringed = Pareto front: no other provider is both at least as accurate and at least as cheap.



Cost & sources

Cost break-even vs Exa

A cheaper price per request only wins if the agent doesn't need proportionally more requests. A provider is cheaper on search than Exa when its **request ratio** (its requests per prompt ÷ Exa's) is below its **price advantage** (Exa's price per request ÷ its own). Requests = searches + page fetches (+ SQL queries for SELECT). Model tokens are excluded here; they are in the ranking.

Provider	Price / 1k requests	Requests / prompt	Price advantage	Request ratio	Search $ / prompt	Search $ / correct answer
Exa	$2.29	20.3	1.00×	1.00×	$0.0463	$0.0772
Parallel	$1.99	15.7	1.15×	0.78×	$0.0314	$0.0515
Keenable search	~$3.88	33.1	0.59×	1.63×	~$0.1283	~$0.2984
Keenable SELECT	~$3.57	24.1	0.64×	1.19×	~$0.0858	~$0.1908
Grok native	$3.02	15.6	0.76×	0.77×	$0.0472	$0.0843
Keenable router	~$3.70	23.1	0.62×	1.14×	~$0.0855	~$0.1781

By prompt type: request ratio vs price advantage

Green = cheaper on search than Exa for that type (request ratio below price advantage).

Type	Parallel	Search	SELECT	Grok	Router
Multi-hop · 23	0.75× / 1.12×	2.08× / 0.85×	1.58× / 0.96×	1.10× / 1.07×	1.48× / 0.89×
List-building · 31	0.87× / 1.14×	1.53× / 0.39×	1.20× / 0.43×	0.71× / 0.68×	1.14× / 0.42×
Regulation · 46	0.60× / 1.00×	1.57× / 0.83×	0.91× / 0.87×	0.69× / 0.70×	0.93× / 0.85×

Sources each provider's answers cite

Domains cited in final answers. "Only here" = domains no other provider cited anywhere in this audit, a rough measure of index coverage. "Wrong, uncited" = wrong answers with no source at all (often answered from memory after search failed).

Provider	Sources / answer	Distinct domains	Only here	Uncited answers	Wrong, uncited	Most cited
Exa	21.0	467	151	0/100	0/16	docs.fcc.gov 51 fcc.gov 39 fcc.report 35 govinfo.gov 32 federalregister.gov 28 faa.gov 23
Parallel	22.3	404	191	0/100	0/12	docs.fcc.gov 56 fcc.gov 43 facebook.com 35 govinfo.gov 30 federalregister.gov 26 satellitetoday.com 21
Keenable search	4.0	122	9	2/100	2/18	docs.fcc.gov 49 fcc.gov 12 ecfr.gov 11 faa.gov 9 govinfo.gov 9 federalregister.gov 7
Keenable SELECT	3.8	116	10	1/100	1/18	docs.fcc.gov 47 ecfr.gov 10 govinfo.gov 10 fcc.gov 9 faa.gov 9 fcc.report 7
Grok native	20.8	434	118	0/100	0/19	docs.fcc.gov 48 fcc.gov 41 govinfo.gov 31 fcc.report 29 federalregister.gov 26 faa.gov 23

Provider	Sources / answer	Distinct domains	Only here	Uncited answers	Wrong, uncited	Most cited
Keenable router	3.7	102	0	2/100	0/13	docs.fcc.gov 48 govinfo.gov 12 fcc.gov 10 ecfr.gov 9 faa.gov 9 law.cornell.edu 6

How we measured

Setup

Prompts	100 · run 2026-10-09
Agent model	`x-ai/grok-4.7` pinned in the audit config
Judge model	`anthropic/claude-sonnet-5.5`
Budget per answer	max tool calls 100, max seconds 1800
Cost of this audit	$366.61

Providers

Exa	OpenRouter openrouter:web_search + openrouter:web_fetch, engine=exa $0.007 per search, $0.001 per fetched page (OpenRouter list price for Exa)
Parallel	OpenRouter openrouter:web_search + openrouter:web_fetch, engine=parallel $0.005 per search, $0.001 per fetched page (OpenRouter list price for Parallel)
Keenable search	Keenable REST /v1/search + /v1/fetch as function tools $0.004 per request (Keenable list price)
Keenable SELECT	Keenable SELECT MCP (https://select.keenable.ai/mcp); the agent writes the SQL the tool describes, plus Keenable fetch $0.004 per request (Keenable list price)
Grok native	OpenRouter openrouter:web_search + openrouter:web_fetch, engine=native Passed through inside the OpenRouter completion charge. No separate list price for native search.
Keenable router	Jev answers one choice, search or select. The agent then has only that tool $0.004 per request (Keenable list price)

CAVEATS

- Every provider ran the same prompts with the same pinned agent model, system prompt and temperature (0.2).
- Budget per run: 100 tool calls and 1800 seconds; runs that hit it were forced to answer and are flagged.
- Exa, Parallel and Grok native ran as OpenRouter server tools (openrouter:web_search and web_fetch), so OpenRouter's own loop decides how often they search inside one model call. Keenable ran as function tools in our own agent loop. The server-tool providers' search cost is what OpenRouter billed, split from the model cost when OpenRouter reported the split, otherwise estimated from list prices (marked ~).
- Keenable search and SELECT are priced at the list price of $0.004 per request. Requests that failed are priced at $0, so an effective rate can sit below list.
- Keenable SELECT is experimental: the agent is given a research skill and writes DuckDB SQL (WEB_SEARCH, SEM_EXTRACT) in the query argument. The other providers do not get that skill. A failed query still counts toward the tool-call budget and is priced at $0, since it is not known whether Keenable bills it.
- Keenable router asks Jev (typesafe/jev-1.13, a small decision model served through OpenRouter) one choice question, search or select, and returns the answer that arm already gave, plus the cost and time of the Jev call. A prompt where either arm failed has no router run and counts as wrong.
- Prompt ids run past 100 with gaps: on 2026-10-09 the 45 prompts that every provider had answered correctly were removed, since they could not separate anyone, and the same number of harder cross-document prompts were added. Ids are never reused.
- Exa: 4 returned no answer, of 100 runs; Grok native: 6 returned no answer, of 100 runs; Keenable search: 1 run errored and 11 hit the budget, of 100 runs; Keenable router: 5 hit the budget, of 98 runs; Keenable SELECT: 1 run errored and 7 hit the budget, of 100 runs. A run that errored or returned no answer counts as wrong; a run that hit the budget was made to answer and graded like any other.
- Judge: anthropic/claude-sonnet-5.5, bundle judge-v3, temperature 0, sees the reference answer.
- One run per prompt is the scored trial. Rerun the audit before treating the ranking as current: providers, models and the web change. About 100 prompts is the smallest set with a real chance of detecting a 10-point accuracy gap. A tie on fewer prompts means the sample was too small to see that gap, not that the providers are equal. The set is 100 prompts because each provider run costs real money; a gap of a few points here is a lead to test at a larger size, not a verdict.
- The judge was not calibrated against hand-graded answers in this audit, so treat small accuracy gaps between providers with care.

Appendix: example answers

Four prompts where providers split most between right and wrong, with each provider's full answer and the judge's reasoning. Same agent model, system prompt and budget for every provider.

EXAMPLE 1 OF 4 · PROMPT 60 · DOMAIN

SpaceX's Gen2 application proposed tracking beacons in the 137-138 MHz and 148-150.05 MHz bands. Did the 1 December 2022 partial grant authorize those beacons, and which direction did the application assign to each band?

Exa **Wrong** 52s · $0.353 · 13 searches, 5 fetches

> *Judge:* The answer reverses the band directions: it puts 137-138 MHz on space-to-Earth and 148-150.05 MHz on Earth-to-space, contradicting facts 3-6 and asserting trap 2. DA-24-222A1 in the DATA states the opposite. It also omits the TT&C distinction and the November 2024 timing (facts 7-9).

No. The 1 December 2022 partial grant did not authorize the beacons. It deferred them.

In FCC 22-91 (adopted 29 November 2022, released 1 December 2022), the Commission granted only up to 7,500 Gen2 satellites in the Ku- and Ka-bands and stated: “We defer consideration of SpaceX’s proposed use of E-band frequencies and tracking beacons.” Source: https://docs.fcc.gov/public/attachments/FCC-22-91A1.txt

The application assigned the usual VHF directions, not the ones printed in that order:

- **137.00–138.00 MHz: space-to-Earth**
- **148.00–150.05 MHz: Earth-to-space**

That assignment is in SpaceX’s 4 October 2022 letter, Exhibit A, and in the 16 December 2022 beacon amendment (SAT-AMD-20221216-00175). The Space Bureau later confirmed both sources in DA 24-1160: beacons “would operate in ... the 137.00-138.00 MHz (space-to-Earth) and 148.00-150.05 MHz (Earth-to-space) ... bands ... as requested in the second amendment,” with downlink channels in 137–138 MHz and uplink channels in 148–150.05 MHz. Source: https://docs.fcc.gov/public/attachments/DA-24-1160A1.txt

FCC 22-91 paragraph 8 reverses those parentheticals (137–138 MHz as Earth-to-space and 148–150.05 MHz as space-to-Earth) while citing the same October 2022 exhibit. That description conflicts with DA 24-1160’s reading of the exhibit and the amendment, and with the U.S. allocations used for Swarm. The formal beacon amendment was filed on 16 December 2022, after the partial grant; before that, the proposal was in the 19 August and 4 October 2022 letters, which the 1 December order treated as part of the pending record and deferred.

Parallel **Wrong** 38s · $0.441 · 5 searches, 2 fetches

> *Judge:* The answer reverses the directions: it puts 137-138 MHz on space-to-Earth and 148-150.05 MHz on Earth-to-space, contradicting facts 3-6 and asserting trap 2. It also dismisses the order's own background paragraph as a drafting error without support, and it does not mention that TT&C during deorbit is a separate grant (fact 7).

No. The FCC’s 1 December 2022 partial grant (FCC 22-91) did not authorize the tracking beacons. It deferred them, along with E-band use, and authorized only up to 7,500 Gen2 satellites in Ku- and Ka-band.
https://docs.fcc.gov/public/attachments/FCC-22-91A1.txt

As described in that order’s beacon discussion (citing SpaceX’s 4 October 2022 letter, Exhibit A), the proposal assigned:

- **137.00–138.00 MHz: space-to-Earth**
- **148.00–150.05 MHz: Earth-to-space**

https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

The same order’s background paragraph reverses those directions. That appears to be a drafting error. The directions above match the ITU allocations and the FCC’s later description of the application, which was granted only on 20

November 2024 (DA 24-1160), not in the 1 December 2022 partial grant. https://docs.fcc.gov/public/attachments/DA-24-1160A1.pdf

Keenable search Correct

28s · $0.092 · 6 searches, 3 fetches

> *Judge:* Core and required facts are correct: beacons deferred, 137 MHz Earth-to-space, 148 MHz space-to-Earth. Context facts 7-9 (TT&C grant distinction, November 2024 later order) are not stated; the answer also adds claims that a later amendment reversed directions, which is supported by the cited notice.

No. The FCC's 1 December 2022 partial grant (FCC 22-91, adopted 29 November 2022) deferred the tracking beacons; it did not authorize them. It authorized only up to 7,500 Gen2 satellites in Ku- and Ka-band at specified shells, and expressly deferred E-band and the beacons. https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

As described in that order, the application assigned **137.00–138.00 MHz to Earth-to-space** and **148.00–150.05 MHz to space-to-Earth**. https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

Later filings reversed those directions (137–138 MHz space-to-Earth and 148–150.05 MHz Earth-to-space), including the 16 December 2022 amendment accepted for filing in January 2023. https://docs.fcc.gov/public/attachments/DOC-390651A1.pdf A March 2024 order still described the December 2022 amendment with the same directions as FCC 22-91, so the FCC's own descriptions of the application's band directions are inconsistent. https://docs.fcc.gov/public/attachments/DA-24-222A1.pdf

Keenable SELECT Correct

143s · $0.056 · 3 searches, 0 fetches

> *Judge:* The core and required facts are all correct. The answer omits the context facts: the separate TT&C deorbit authority (7), the November 2024 order that later addressed beacons (8), and the warning not to conflate that order with the 2022 one (9).

No. The 1 December 2022 partial grant did not authorize the tracking beacons. FCC 22-91 (adopted 29 November 2022, released 1 December 2022) granted in part and deferred in part SpaceX's Gen2 application, and it expressly deferred the beacons: "We defer consideration of SpaceX's proposed use of E-band frequencies and tracking beacons." https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

As the order describes the application, the directions are reversed from the usual VHF satellite allocation:

- **137.00–138.00 MHz: Earth-to-space**
- **148.00–150.05 MHz: space-to-Earth**

https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

A later SpaceX amendment (SAT-AMD-20221216-00175), accepted for filing after the partial grant, proposed the opposite directions—137–138 MHz space-to-Earth and 148–150.05 MHz Earth-to-space. That is not what the Gen2 application, as recited in the 1 December 2022 order, assigned to each band. https://docs.fcc.gov/public/attachments/DOC-390651A1.pdf

Grok native Wrong

68s · $0.652 · 12 searches, 3 fetches

> *Judge:* The answer correctly says the beacons were not authorized and cites the para 1 deferral. It claims para 68 gives the reverse direction pairing (137 MHz space-to-Earth) and calls that the more specific description, which the data does not support, so facts 3-6 are contradicted. It also omits facts 7-9, including the November 2024 order.

No. The 1 December 2022 partial grant did not authorize those beacons.

In FCC 22-91 (adopted 29 November 2022, released 1 December 2022), the Commission granted SpaceX authority for up to 7,500 Gen2 satellites in Ku- and Ka-band and expressly deferred the tracking/emergency beacons, along with E-band. It also said it was deferring any action on SpaceX's described beacon use pending further review. Source: https://docs.fcc.gov/public/attachments/FCC-22-91A1.txt (paras. 1, 69).

On direction, that same order is internally inconsistent:

- Paragraph 68, citing SpaceX's 4 October 2022 letter, Exhibit A at 1, says the beacons would use **137.00–138.00 MHz space-to-Earth** and **148.00–150.05 MHz Earth-to-space**.

- Paragraph 8, citing the 19 August 2022 letter and the same October letter more generally, reverses that: **137.00–138.00 MHz Earth-to-space** and **148.00–150.05 MHz space-to-Earth**.

The Exhibit A citation in paragraph 68 is the more specific description of what SpaceX had put on the record before the grant. At that time SpaceX also said it was not seeking U.S. market access for the beacons and intended to use them only outside the United States. A formal amendment seeking beacon authority (SAT-AMD-20221216-00175) was filed on 16 December 2022, after this partial grant. A later FCC order, DA 24-222, describes that amendment with the reversed pairing used in paragraph 8, so the Commission's own summaries do not agree. Source for the deferral and the later description: https://docs.fcc.gov/public/attachments/DA-24-222A1.pdf.

Keenable router **Correct**

143s · $0.056 · 3 searches, 0 fetches

> *Judge:* The core and required facts are all correct. The answer omits the context facts: the separate TT&C deorbit authority (7), the November 2024 order that later addressed beacons (8), and the warning not to conflate that order with the 2022 one (9).

No. The 1 December 2022 partial grant did not authorize the tracking beacons. FCC 22-91 (adopted 29 November 2022, released 1 December 2022) granted in part and deferred in part SpaceX's Gen2 application, and it expressly deferred the beacons: "We defer consideration of SpaceX's proposed use of E-band frequencies and tracking beacons." https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

As the order describes the application, the directions are reversed from the usual VHF satellite allocation:

- **137.00–138.00 MHz: Earth-to-space**
- **148.00–150.05 MHz: space-to-Earth**

https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf

A later SpaceX amendment (SAT-AMD-20221216-00175), accepted for filing after the partial grant, proposed the opposite directions—137–138 MHz space-to-Earth and 148–150.05 MHz Earth-to-space. That is not what the Gen2 application, as recited in the 1 December 2022 order, assigned to each band. https://docs.fcc.gov/public/attachments/DOC-390651A1.pdf

EXAMPLE 2 OF 4 · PROMPT 50 · DOMAIN

La décision Arcep n° 2022-1102 du 25 mai 2022 autorisait-elle déjà les stations terriennes en mouvement? Quelle décision a ajouté les ESIM, pour quel seul système UIT, et quelle PIRE maximale leur est imposée?

Exa Correct

13s · $0.068 · 4 searches, 0 fetches

> *Judge:* The core and required facts are all correct. Only the context facts 6 to 9 are missing: the 2023 bands, the no-interference-protection condition, the exclusion of STEAM-1B, and the consultation date discrepancy.

Non. La décision n° 2022-1102 du 25 mai 2022 n'autorisait que des stations terriennes fixes (terminaux résidentiels) avec le système non géostationnaire enregistré à l'UIT sous le nom STEAM-1, dans les bandes 10,95-12,70 GHz (espace vers Terre) et 14-14,5 GHz (Terre vers espace), avec une PIRE maximale de 60 dBW pour ces stations fixes. Elle ne visait pas les stations terriennes en mouvement.

Les ESIM ont été ajoutées par la décision n° 2023-0680 du 28 mars 2023, qui modifie la décision n° 2022-1102. À ce stade, elles n'étaient autorisées que pour le seul système UIT « STEAM-1 ». La PIRE maximale imposée à ces ESIM est de 54,5 dBW (conditions de la décision ECC/DEC/(18)05).

Sources : https://www.arcep.fr/uploads/tx_gsavis/22-1102.pdf et https://www.arcep.fr/uploads/tx_gsavis/23-0680.pdf

Parallel Correct

19s · $0.074 · 4 searches, 3 fetches

> *Judge:* The core and required facts are all correct: ESIM were not in the 2022 decision, decision 2023-0680 added them, the only system is STEAM-1, and the maximum EIRP is 54.5 dBW. Only the context facts are missing: the frequency bands, the no-interference-protection condition, the later STEAM-1B decision, and the consultation date discrepancy. The 54.5 dBW wording in the DATA window comes from decision 2023-0679 (OneWeb), not 2023-0680.

Non. La décision n° 2022-1102 du 25 mai 2022 n'autorisait que des stations terriennes fixes (terminaux résidentiels) avec le système non géostationnaire « STEAM-1 », et non des stations en mouvement.

C'est la décision n° 2023-0680 du 28 mars 2023 qui a modifié la n° 2022-1102 pour y ajouter les ESIM (stations terriennes en mouvement installées sur des plateformes mobiles). L'annexe limite ces liaisons au seul système enregistré à l'UIT sous le nom « STEAM-1 ». La PIRE maximale imposée à ces ESIM est de 54,5 dBW (conditions de la décision ECC/DEC/(18)05).

Sources : [décision n° 2022-1102 (PDF)](https://www.arcep.fr/uploads/tx_gsavis/22-1102.pdf) ; [décision n° 2023-0680 (PDF)](https://www.arcep.fr/uploads/tx_gsavis/23-0680.pdf) ; [communiqué Arcep du 2 juin 2022, mis à jour](https://www.arcep.fr/actualites/actualites-et-communiques/detail/n/frequences-020622.html).

Keenable search Correct

87s · $0.191 · 16 searches, 3 fetches

> *Judge:* Les faits clés (1 à 4) sont corrects. Les faits de contexte 6 à 9 ne sont pas mentionnés (bandes de 2023, absence de garantie de non-brouillage, STEAM-1B ultérieur, écart de dates de consultation). Le fait 5 est couvert, avec des fréquences de 10,95-12,70 GHz non précisées pour la liaison descendante mais citées comme 10,95-12,70 et 14-14,5 GHz.

Non. La décision n° 2022-1102 du 25 mai 2022 n'autorisait que des stations terriennes fixes, pour un accès internet fixe par satellite, dans les bandes 10,95-12,70 GHz et 14-14,5 GHz, en application de la décision ECC/DEC/(17)04. Elle ne visait pas les stations terriennes en mouvement.
https://www.arcep.fr/uploads/tx_gsavis/22-1102.pdf

C'est la décision n° 2023-0680 du 28 mars 2023, qui modifie la décision n° 2022-1102, qui a ajouté les ESIM. Elles ne sont autorisées qu'avec le seul système non géostationnaire enregistré à l'UIT sous le nom « STEAM-1 ». Leur PIRE maximale est de 54,5 dBW, conformément à la décision ECC/DEC/(18)05.
https://www.arcep.fr/uploads/tx_gsavis/23-0680.pdf

Keenable SELECT Wrong

175s · $0.110 · 5 searches, 2 fetches

> *Judge:* The answer cites decision 2024-2687 (Dec 2024, STEAM-1B) instead of 2023-0680 of 28 March 2023 with STEAM-1, so facts 2, 3, 6 and 8 are contradicted and trap 2 (STEAM-1B) is asserted. The 54.5 dBW PIRE and the non-

> authorization of ESIM in 2022 are correct; non-interference and consultation dates are missing.

Non. La décision Arcep n° 2022-1102 du 25 mai 2022 n'autorisait que des **stations terriennes fixes** (terminaux résidentiels) avec le système UIT « STEAM-1 », dans les bandes 10,95-12,70 GHz et 14-14,5 GHz. Elle ne visait pas les stations terriennes en mouvement.
Source : https://www.arcep.fr/uploads/tx_gsavis/22-1102.pdf

Les ESIM ont été ajoutées par la décision n° 2024-2687 du 19 décembre 2024, qui modifie la n° 2022-1102. Après restriction de la demande de Starlink (initialement STEAM-1B et USASAT-NGSO-3X), l'autorisation ne porte que sur **un seul système UIT : « STEAM-1B »**, pour des liaisons avec des stations terriennes fixes et des ESIM (bandes 10,70-12,75 GHz et 14,0-14,50 GHz).
Source : https://www.arcep.fr/uploads/tx_gsavis/24-2687.pdf
Communiqué : https://en.arcep.fr/news/press-releases/view/n/frequencies-191224.html

PIRE maximale imposée aux ESIM : 54,5 dBW, par renvoi aux conditions de la décision ECC/DEC/(18)05 (annexe de la décision 2024-2687).
Source : https://www.arcep.fr/uploads/tx_gsavis/24-2687.pdf

Grok native Correct

14s · $0.109 · 3 searches, 1 fetches

> *Judge:* The core and required facts are all correct. Only context facts are missing: the frequency bands of the 2023 decision, the no-interference-protection clause, the STEAM-1B exclusion and the consultation date discrepancy.

Non. La décision n° 2022-1102 du 25 mai 2022 n'autorisait que des stations terriennes fixes (chez des clients résidentiels), avec le système non géostationnaire enregistré à l'UIT sous le nom STEAM-1, et une PIRE maximale de 60 dBW pour ces stations fixes. Elle ne visait pas les stations terriennes en mouvement.

C'est la décision n° 2023-0680 du 28 mars 2023, qui modifie la décision n° 2022-1102, qui a ajouté les ESIM. Dans son annexe, le seul système UIT autorisé pour les liaisons avec les stations fixes et les ESIM est « STEAM-1 ». La PIRE maximale imposée aux ESIM est de 54,5 dBW (conditions de la décision ECC/DEC/(18)05).

Sources : [décision n° 2022-1102 (PDF)](https://www.arcep.fr/uploads/tx_gsavis/22-1102.pdf), [décision n° 2023-0680 (PDF)](https://www.arcep.fr/uploads/tx_gsavis/23-0680.pdf), [communiqué Arcep] (https://www.arcep.fr/actualites/actualites-et-communiques/detail/n/frequences-020622.html).

Keenable router Wrong

175s · $0.110 · 5 searches, 2 fetches

> *Judge:* The answer cites decision 2024-2687 (Dec 2024, STEAM-1B) instead of 2023-0680 of 28 March 2023 with STEAM-1, so facts 2, 3, 6 and 8 are contradicted and trap 2 (STEAM-1B) is asserted. The 54.5 dBW PIRE and the non-authorization of ESIM in 2022 are correct; non-interference and consultation dates are missing.

Non. La décision Arcep n° 2022-1102 du 25 mai 2022 n'autorisait que des **stations terriennes fixes** (terminaux résidentiels) avec le système UIT « STEAM-1 », dans les bandes 10,95-12,70 GHz et 14-14,5 GHz. Elle ne visait pas les stations terriennes en mouvement.
Source : https://www.arcep.fr/uploads/tx_gsavis/22-1102.pdf

Les ESIM ont été ajoutées par la décision n° 2024-2687 du 19 décembre 2024, qui modifie la n° 2022-1102. Après restriction de la demande de Starlink (initialement STEAM-1B et USASAT-NGSO-3X), l'autorisation ne porte que sur **un seul système UIT : « STEAM-1B »**, pour des liaisons avec des stations terriennes fixes et des ESIM (bandes 10,70-12,75 GHz et 14,0-14,50 GHz).
Source : https://www.arcep.fr/uploads/tx_gsavis/24-2687.pdf
Communiqué : https://en.arcep.fr/news/press-releases/view/n/frequencies-191224.html

PIRE maximale imposée aux ESIM : 54,5 dBW, par renvoi aux conditions de la décision ECC/DEC/(18)05 (annexe de la décision 2024-2687).
Source : https://www.arcep.fr/uploads/tx_gsavis/24-2687.pdf

EXAMPLE 3 OF 4 · PROMPT 98 · WIDE

DA-26-36 authorizes Gen2 W-band operations in 92.0-94.0, 94.1-95.0, 95.0-100.0, 102.0-109.5, and 111.8-114.25 GHz. Are those operations a protected downlink, and what minimum earth-station elevation does the order require at each satellite altitude?

Exa Correct

38s · $0.190 · 6 searches, 1 fetches

> *Judge:* All checklist facts are stated correctly and no trap is asserted. The 10/20/25 degree angles come from the amendment text in the DATA window, and the unprotected Earth-to-space waiver basis is consistent with the DA-26-36 text.

No. Those Gen2 W-band operations are not a protected downlink. DA-26-36 authorizes them as Earth-to-space (uplink) operations, and only on an unprotected, non-interference basis.

The grant lists 92.0–94.0, 94.1–95.0, 95.0–100.0, 102.0–109.5, and 111.8–114.25 GHz as Earth-to-space and grants a waiver of the U.S. Table for NGSO uplink use in those bands. Condition 7 requires SpaceX to operate on an unprotected, non-interference basis. The order also says the operations are non-conforming with the International Table and are unprotected under ITU Radio Regulations Article 4.4. Satellites use the bands on a receive-only basis (uplink reception), mainly for parabolic earth-station backhaul—not as a protected space-to-Earth downlink.

In the waiver discussion (Order ¶ 11), the authorized W-band earth stations have these minimum elevation angles:

- Below 400 km: 10 degrees
- 400–500 km: 20 degrees
- Above 500 km: 25 degrees

Those angles are stated as the operating parameters of the operations being granted, not as a separately numbered grant condition. The grant also requires operations to follow the applicant's technical specifications and commitments unless the order says otherwise.

Source: https://docs.fcc.gov/public/attachments/DA-26-36A1.txt

Parallel Correct

23s · $0.119 · 2 searches, 1 fetches

> *Judge:* All core and required facts are stated correctly. Only the context fact about footnote 21 and ITU Article 4.4 non-conformance is missing; no traps asserted.

No. Those bands are not a protected downlink.

In DA-26-36 (released January 9, 2026), the Space Bureau authorizes SpaceX Gen2 W-band use of 92.0–94.0, 94.1–95.0, 95.0–100.0, 102.0–109.5, and 111.8–114.25 GHz as **Earth-to-space uplinks**, on a **receive-only** basis at the satellite, and only on an **unprotected, non-interference** basis. The protected-downlink characterization does not apply. See grant condition 7 and Order ¶¶ 9 and 11: https://docs.fcc.gov/public/attachments/DA-26-36A1.txt

The same paragraph states the minimum W-band earth-station elevation angles the Bureau relies on for the waiver:

- **Below 400 km:** 10 degrees
- **400–500 km:** 20 degrees
- **Above 500 km:** 25 degrees

Those angles are SpaceX commitments incorporated into the grant ("operations under this grant must comport with the legal and technical specifications and commitments set forth by the applicant").

Keenable search Correct

20s · $0.062 · 4 searches, 2 fetches

> *Judge:* All core and required facts are correct. Only the context fact about Footnote 21 (non-conforming under ITU RR Article 4.4) is not stated in the answer.

No. Those W-band operations are not a protected downlink. DA-26-36 authorizes them as Earth-to-space (uplink) operations under a waiver, on an unprotected, non-interference basis. The order also states that SpaceX will use the W-

band frequencies on a receive-only basis at the satellite.

The same order requires these minimum W-band earth-station elevation angles:

- **10°** for satellites operating below 400 km
- **20°** for satellites operating between 400 km and 500 km
- **25°** for satellites operating above 500 km

Source: https://docs.fcc.gov/public/attachments/DA-26-36A1.pdf (also https://docs.fcc.gov/public/attachments/DA-26-36A1_Rcd.pdf)

Keenable SELECT **Wrong** 232s · $0.088 · 5 searches, 0 fetches

> *Judge:* All facts are stated, including the 10/20/25 degree elevation angles. The answer asserts that the order sets no elevation requirement, only recites SpaceX's proposal, which matches the trap and conflicts with the expected answer that the order requires these angles.

No. Those operations are not a protected downlink.

DA-26-36 authorizes them as Earth-to-space NGSO uplinks, and only under a waiver of the U.S. Table of Frequency Allocations. Paragraph 7 says SpaceX must operate in 92.0–94.0, 94.1–95.0, 95.0–100.0, 102.0–109.5, and 111.8–114.25 GHz on an unprotected, non-interference basis. Paragraph 9 repeats that these W-band operations are non-conforming with the International Table and are unprotected and non-interference. Source: https://docs.fcc.gov/public/attachments/DA-26-36A1.pdf

The order does not impose a minimum earth-station elevation as a grant condition. Paragraph 11 only recites SpaceX's proposal: receive-only use of these frequencies, with earth stations at a minimum elevation of 10° below 400 km, 20° between 400 and 500 km, and 25° above 500 km. The operative W-band condition in paragraph 7 requires only unprotected, non-interference operation.

Grok native **Correct** 15s · $0.126 · 2 searches, 1 fetches

> *Judge:* Only fact 3 (footnote 21 on non-conformance with the International Table under ITU RR Article 4.4) is missing; all other facts are correct and no traps were asserted. The DATA window does not show the elevation-angle text, but the answer's elevation figures and uplink and unprotected claims match the checklist.

No. Those W-band operations are not a protected downlink. DA-26-36 authorizes them as Earth-to-space uplinks on an unprotected, non-interference basis.

The grant lists 92.0–94.0, 94.1–95.0, 95.0–100.0, 102.0–109.5, and 111.8–114.25 GHz as Earth-to-space. Grant condition 7 requires SpaceX to operate in those bands on an unprotected and non-interference basis. The order describes the use as satellite receive-only (uplink), mainly for parabolic earth-station backhaul, not as a protected space-to-Earth downlink.

The order states that W-band earth stations will use these minimum elevation angles:

- **Below 400 km:** 10 degrees
- **400–500 km:** 20 degrees
- **Above 500 km:** 25 degrees

Source: https://docs.fcc.gov/public/attachments/DA-26-36A1.txt

Keenable router **Wrong** 233s · $0.088 · 5 searches, 0 fetches

> *Judge:* All facts are stated, including the 10/20/25 degree elevation angles. The answer asserts that the order sets no elevation requirement, only recites SpaceX's proposal, which matches the trap and conflicts with the expected answer that the order requires these angles.

No. Those operations are not a protected downlink.

DA-26-36 authorizes them as Earth-to-space NGSO uplinks, and only under a waiver of the U.S. Table of Frequency Allocations. Paragraph 7 says SpaceX must operate in 92.0–94.0, 94.1–95.0, 95.0–100.0, 102.0–109.5, and 111.8–114.25 GHz on an unprotected, non-interference basis. Paragraph 9 repeats that these W-band operations are non-conforming

with the International Table and are unprotected and non-interference. Source: https://docs.fcc.gov/public/attachments/DA-26-36A1.pdf

The order does not impose a minimum earth-station elevation as a grant condition. Paragraph 11 only recites SpaceX's proposal: receive-only use of these frequencies, with earth stations at a minimum elevation of 10° below 400 km, 20° between 400 and 500 km, and 25° above 500 km. The operative W-band condition in paragraph 7 requires only unprotected, non-interference operation.

If destroying the vehicle would put more people at risk than letting it fly, do the flight-safety limits still require an abort, and does a lost track force one?

Exa **Correct** 29s · $0.107 · 4 searches, 2 fetches

> *Judge:* All core and required facts are correct. The answer omits the context facts: the abort on a valid-data violation (7), the abort for a compromised flight safety system (8), the 450.145 and 450.143 consequence thresholds (9-11), and the alternative debris-containment option (14).

No. Under FAA flight-abort rules, limits must not require an abort that raises collective public risk, and a lost track does not by itself force an immediate abort.

Abort that increases risk. 14 CFR § 450.108(d)(6) requires flight safety limits that are "designed to avoid flight abort that results in increased collective risk to the public in uncontrolled areas, compared to continued flight." FAA Advisory Circular 450.108-1 § 10.6 treats that as a requirement that abort not present higher public risk than continued flight. Limits are placed so the lower-risk choice is used—abort or continued flight—including when the vehicle has left the limits of a useful mission (§ 450.108(c)(2) and AC § 9.3). Separately, § 450.108(e) says a flight need not be aborted to protect against high-consequence events in uncontrolled areas once critical parameters are validated, the vehicle can still achieve a useful mission, abort would not materially decrease that risk, and there are no key flight safety events.

Lost track. Invalid tracking data does not automatically require abort. § 450.108(f)(2)(iii) requires abort only "in accordance with methods used to satisfy (d)(3)" when tracking data is invalid **and** further flight has the potential to violate a flight safety limit. Section 450.108(d)(3) only requires that the limits account for possible loss of the data needed to evaluate abort rules. The accepted approach in AC 450.108-1 § 10.3 is a precomputed Duration of Acceptable Data Loss (also called data-loss flight time or "green numbers"): if all tracking data are lost, a countdown starts, and abort is taken when it reaches zero—not at the instant of track loss. If further flight has no potential to violate a flight safety limit, (f)(2)(iii) does not require abort. A valid-data limit violation under (f)(2)(i) also does not apply while the track is invalid.

Sources: [14 CFR § 450.108](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108); [AC 450.108-1 (July 27, 2021)](https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_450.108-1_Flight_Abort_Rule_Development.pdf).

Parallel **Correct** 31s · $0.150 · 5 searches, 2 fetches

> *Judge:* The answer gets all the core and required points right (avoid abort that raises collective risk, lost track not an automatic abort, data-loss abort tied to potential limit violation). It omits the context facts 7-14: the other abort triggers, the 450.143/450.145 consequence thresholds, end-of-flight abort, and the 450.145 alternative.

I'll look up current flight-safety rules on abort versus continued flight when termination would increase risk, and on lost-track requirements.No. Under FAA launch and reentry rules, flight-safety limits must not require an abort that raises public risk, and a lost track does not by itself force an immediate abort.

Abort when destruction is riskier. Flight safety limits must be designed to avoid a flight abort that increases collective risk to the public in uncontrolled areas compared with continued flight. [14 CFR § 450.108(d)(6)](https://www.ecfr.gov/current/title-14/section-450.108). FAA guidance states that abort action should not present a higher risk than not aborting, and that limits should be placed so the lower-risk choice is used. [AC 450.108-1, ¶¶ 9.3 and 10.6](https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_450.108-1_Flight_Abort_Rule_Development.pdf).

Lost track. Invalid tracking data requires abort only under the operator's pre-established data-loss method, and only if further flight still has the potential to violate a flight safety limit. [14 CFR § 450.108(f)(2)(iii)](https://www.ecfr.gov/current/title-14/section-450.108); limits must account for loss of the data needed to evaluate abort rules. [§ 450.108(d)(3)](https://www.ecfr.gov/current/title-14/section-450.108). The accepted means of compliance uses a duration of acceptable data loss (a countdown), not an automatic immediate destruct. Abort on data loss is not required once the vehicle can no longer reach a flight safety limit (including after orbital insertion). If the hazard footprint is already over an uncontrolled populated area, termination is delayed rather than taken immediately, and if the vehicle is still inside the normal envelope the allowed data-loss time should cover reasonably expected dropouts. [AC 450.108-1, ¶¶ 10.3 and 12.2.3](https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_450.108-1_Flight_Abort_Rule_Development.pdf).

Keenable search **Correct** 21s · $0.055 · 3 searches, 2 fetches

> *Judge:* The core and required facts are all correct. Only context facts are missing: the abort on valid data showing a limit violation (7), the abort on approaching a compromising vehicle state (8), the 450.145 and 450.143 flight safety system thresholds (9-11), and the alternative debris-containment option (14).

Under FAA 14 CFR § 450.108, no. Limits must be drawn so abort is not required when it would raise public risk, and invalid tracking forces abort only if further flight could still violate a flight safety limit.

Abort that increases risk. Flight safety limits must be "designed to avoid flight abort that results in increased collective risk to the public in uncontrolled areas, compared to continued flight" ([14 CFR § 450.108(d)(6)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). FAA AC 450.108-1 (July 27, 2021) states the same point: abort action should not present a higher risk to the public than not aborting, and that comparison can change where the limits are placed ([AC 450.108-1](https://faraim.online/advisory-circulars/ac-450-108-1)). Related rules: abort is not required against high-consequence events once critical parameters are validated, the vehicle can still achieve a useful mission, abort would not materially decrease that risk, and there are no key flight safety events ([§ 450.108(e)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). Limits must also keep conditional expected casualties from abort (or from flight outside a useful mission) at or below \(1 \times 10^{-2}\) in uncontrolled areas ([§ 450.108(c)(4)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)).

Lost track. Invalid tracking does not by itself force an abort. The flight safety system must abort "if tracking data is invalid and further flight has the potential to violate a flight safety limit," using methods that already account for loss of valid data ([§ 450.108(f)(2)(iii)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108) and [(d)(3)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). AC 450.108-1 treats allowable data-loss duration as something the operator calculates (normal envelope, populated overflight, unintended trajectory), not as an automatic destruct on loss of track.

This is the FAA commercial launch/reentry rule. Military range procedures can differ and are not covered here.

Keenable SELECT **Wrong** 253s · $0.086 · 4 searches, 0 fetches

> *Judge:* The answer covers the core facts (collective-risk limit, lost track not forcing abort, conditional abort on invalid data). It omits the other abort triggers, the 450.145/450.143 consequence thresholds, and the provisions for ending abort protection or substituting a 450.145 system. It also never states that an abort is not required merely because it is available.

No. Under current FAA rules, flight-safety limits must not force an abort that raises public risk, and a lost track forces an abort only if further flight could still violate a limit.

Abort when destruction is riskier. [14 CFR § 450.108(d)(6)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108) requires flight safety limits that "avoid flight abort that results in increased collective risk to the public in uncontrolled areas, compared to continued flight." [AC 450.108-1](https://faraim.online/advisory-circulars/ac-450-108-1) states the same point: abort action should not present a higher risk than not aborting, and termination should be delayed if uncontrolled populated areas would be hazarded. That is also how range safety has been applied in practice: on a 2018 Ariane 5 that veered toward Kourou, the range safety officer did not terminate because debris could hit the adjacent town ([Range safety](https://en.wikipedia.org/wiki/Range_safety)).

Lost track. A lost track does **not** by itself force an abort. [§ 450.108(f)(2)(iii)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108) requires abort only if tracking data is invalid **and** further flight has the potential to violate a flight safety limit. Limits must account for loss of the data needed to evaluate the abort rules ([§ 450.108(d)(3)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). AC 450.108-1 describes one compliant method: a predefined data-loss countdown, with abort when the timer reaches zero—and delay if abort would hazard an uncontrolled populated area.

Older range rules were stricter. Under EWR 127-1, complete loss of both required tracking sources was treated as a prima facie reason to terminate even with no sign the vehicle was off course; loss of only one source was not ([National Academies, *Streamlining Space Launch Range Safety*, ch. 4](https://nap.nationalacademies.org/read/9790/chapter/6)). That is not the current FAA rule.

Grok native **Wrong** 29s · $0.104 · 4 searches, 3 fetches

> *Judge:* The answer covers the core points on risk-increasing aborts, lost track and invalid-data aborts, but never says an abort is not required merely because it is available (core fact 3). It also omits the compromised-FSS abort condition, the 450.143/450.145 consequence thresholds, and the provisions on ending abort protection and the alternative to (c)(2) and (c)(4).

I'll look up the flight-safety rules on abort versus risk, and whether a lost track forces an abort.The FAA flight-abort rule looks like the right source. I'll read the regulation and the advisory circular on risk and lost track.No. Under the FAA commercial launch and reentry flight-abort rules in 14 CFR § 450.108, flight-safety limits are not supposed to command an abort when destroying the vehicle would raise public risk, and a lost track does not by itself force an immediate abort.

Abort that increases risk. Limits must be designed to avoid a flight abort that increases collective risk to the public in uncontrolled areas compared with continued flight. That is a design constraint on where the limits are drawn, not a real-time waiver after a limit is already violated. If valid real-time data show a limit has been crossed, the flight safety system must abort. FAA guidance treats this as a common-sense rule: abort action should not present a higher public risk than not aborting, and the comparison is between abort and continued flight (including conditional expected casualties).

Lost track. Invalid tracking data forces an abort only under the methods used to account for loss of valid data, and only if further flight has the potential to violate a flight safety limit. It is not an automatic destruct the moment the track is lost. FAA guidance describes a Duration of Acceptable Data Loss: if all tracking data are lost while the abort system is active, a pre-set countdown starts from the vehicle state at loss of data, and abort action is taken when that timer reaches zero. The uncertainty is that data loss may be benign (sensor or link drop, breakup where abort would do nothing) or may accompany an off-nominal trajectory, so the rule is a timed criterion, not an instant one.

Sources: [14 CFR § 450.108](https://www.law.cornell.edu/cfr/text/14/450.108) (especially (d)(3), (d)(6), and (f)(2)); [AC 450.108-1, Flight Abort Rule Development](https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_450.108-1_Flight_Abort_Rule_Development.pdf) (data-loss and “abort should never increase risk” guidance). This is the FAA Part 450 rule set, not a general aviation or military range-safety standard.

Keenable router **Correct** 21s · $0.055 · 3 searches, 2 fetches

> *Judge:* The core and required facts are all correct. Only context facts are missing: the abort on valid data showing a limit violation (7), the abort on approaching a compromising vehicle state (8), the 450.145 and 450.143 flight safety system thresholds (9-11), and the alternative debris-containment option (14).

Under FAA 14 CFR § 450.108, no. Limits must be drawn so abort is not required when it would raise public risk, and invalid tracking forces abort only if further flight could still violate a flight safety limit.

Abort that increases risk. Flight safety limits must be “designed to avoid flight abort that results in increased collective risk to the public in uncontrolled areas, compared to continued flight” ([14 CFR § 450.108(d)(6)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). FAA AC 450.108-1 (July 27, 2021) states the same point: abort action should not present a higher risk to the public than not aborting, and that comparison can change where the limits are placed ([AC 450.108-1](https://faraim.online/advisory-circulars/ac-450-108-1)). Related rules: abort is not required against high-consequence events once critical parameters are validated, the vehicle can still achieve a useful mission, abort would not materially decrease that risk, and there are no key flight safety events ([§ 450.108(e)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). Limits must also keep conditional expected casualties from abort (or from flight outside a useful mission) at or below 1×10^{-2} in uncontrolled areas ([§ 450.108(c)(4)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)).

Lost track. Invalid tracking does not by itself force an abort. The flight safety system must abort “if tracking data is invalid and further flight has the potential to violate a flight safety limit,” using methods that already account for loss of valid data ([§ 450.108(f)(2)(iii)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108) and [(d)(3)](https://www.ecfr.gov/current/title-14/chapter-III/subchapter-C/part-450/subpart-C/section-450.108)). AC 450.108-1 treats allowable data-loss duration as something the operator calculates (normal envelope, populated overflight, unintended trajectory), not as an automatic destruct on loss of track.

This is the FAA commercial launch/reentry rule. Military range procedures can differ and are not covered here.